

## 2.2.26

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### Question

Find the angle between the pair of lines given by

$$\vec{\mathbf{r}} = 3\hat{i} + 2\hat{j} - 4\hat{k} + \lambda(\hat{i} + 2\hat{j} + 2\hat{k})$$

$$\text{and } \vec{\mathbf{r}} = 2\hat{j} + \mu(3\hat{i} + 2\hat{j} + 6\hat{k})$$

**solution:**

Let

$$\mathbf{A} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \mathbf{B} = \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix} \quad (1)$$

The angle between the lines equals the angle between a and b

$$\cos \theta = \frac{\mathbf{a}^T \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} \quad (2)$$

Compute the pieces:

$$\mathbf{a}^T \mathbf{b} = \begin{pmatrix} 1 & 2 & 3 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \\ 6 \end{pmatrix} \quad (3)$$

$$\mathbf{a}^T \mathbf{b} = 1.3 + 2.2 + 2.6 = 9 \quad (4)$$

Thus,

$$\cos \theta = \frac{19}{3.7} = \frac{19}{21} \quad (5)$$

Then:

$$\theta = \cos^{-1} \left( \frac{19}{21} \right) \quad (6)$$

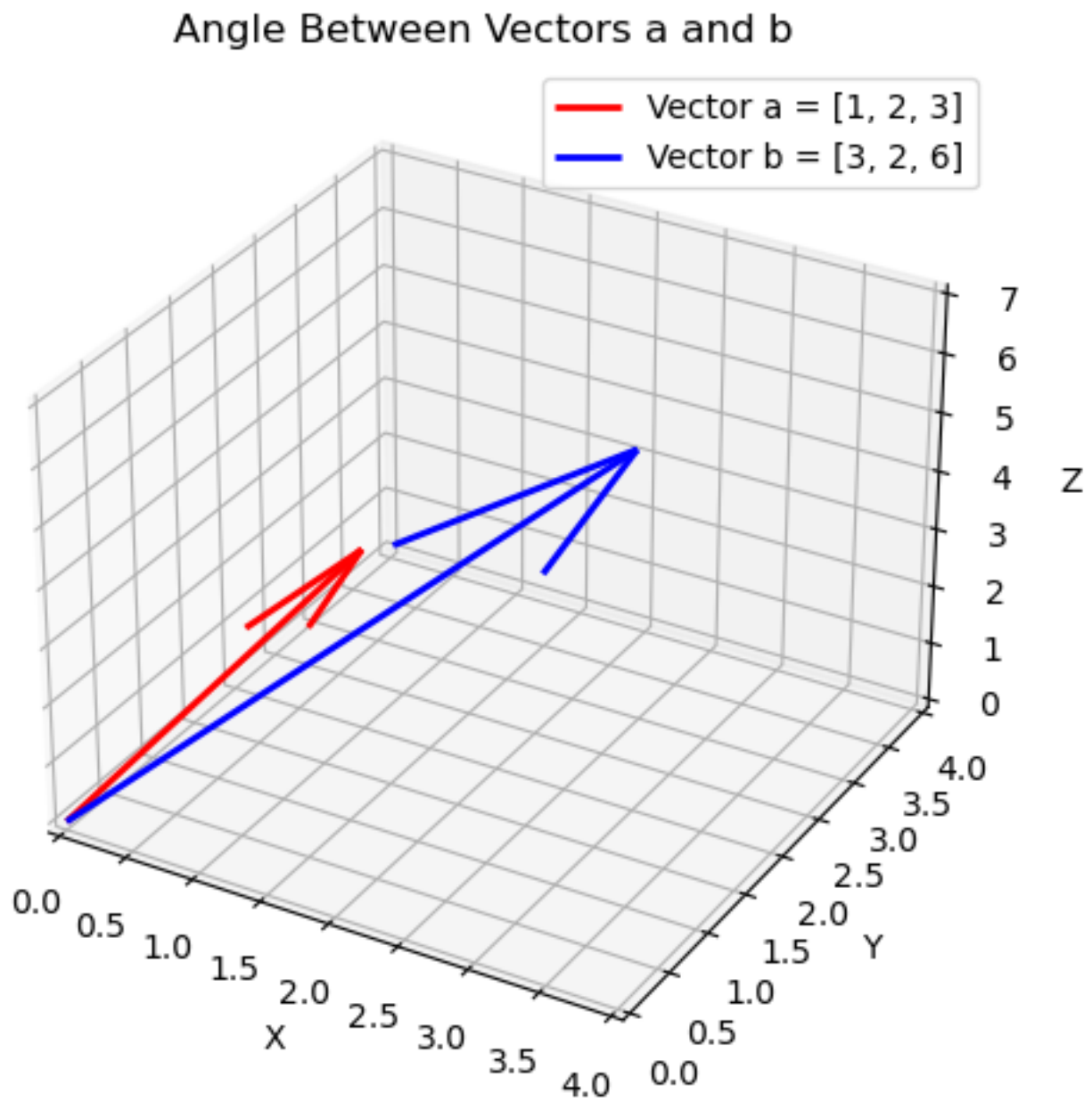


Fig. 1